Optimizing RAG Parameters and Comparing CPU and GPU Performance for the LendSuri Online Lending Platform Chatbot

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Abstract—This group experiment evaluates three character-based chunking configurations for LendSuri, a retrieval-augmented generation (RAG) chatbot answering questions about Philippine online lending platforms using researcher-prepared SEC-related records. Three unchanged member PDFs (379 records) and nine fixed questions were tested with a High-RAM Google Colab CPU and an NVIDIA L4 GPU. The system used all-MiniLM-L6-v2 embeddings, Chroma retrieval (top-k = 3), and Llama-SEA-LION-v3-8B-IT. On the L4 GPU, the 1,200-character/200-overlap setting answered 8 of 9 questions correctly, with a mean response time of 10.87 s and 2.12 s indexing time. Its recorded response-time speedup relative to High-RAM CPU was 44.72 times. A default free-tier CPU did not finish an answer after approximately 40 minutes. Results use two measured repetitions per question; the report distinguishes the later conversation-history interface from the hardware benchmarks.

Keywords—Retrieval-augmented generation, LendSuri, online lending platforms, ChromaDB, SEA-LION, CPU, GPU, chunk size, latency.

I. INTRODUCTION

LendSuri assists users in finding and interpreting dated Securities and Exchange Commission (SEC)-related evidence about online lending platforms (OLPs) in the Philippines. It is an academic information-retrieval prototype, not a current licensing or legitimacy certification service. Following the individual PT-F2 chunking experiments, the group combined all three members' unchanged datasets to compare RAG settings and processing performance under CPU and GPU execution [1], [2].

This experiment examined how three previously selected chunk settings perform on the same corpus and question set, and whether GPU execution offers a practical speed advantage without sacrificing grounded answer quality.

II. METHODOLOGY

A. Corpus and Fixed Questions

The knowledge base contained 379 text-extractable pages/records from the same three member PDFs used in the earlier activity: Member 1 (127 pages), Member 2 (126), and Member 3 (126). The documents are researcher-prepared records referencing dated SEC registry snapshots, historical listings, and advisories; they were neither merged nor rewritten for this experiment [2]. Nine questions were retained: three per member, with three Easy, three Average, and three Difficult items addressing entity identification, historical comparisons, and potentially misleading social-media impersonation claims [3].

B. RAG Implementation and Variables

The Google Colab implementation used PyPDFLoader, RecursiveCharacterTextSplitter, sentence-transformers/all-MiniLM-L6-v2 embeddings, a Chroma vector store, and the aisingapore/Llama-SEA-LION-v3-8B-IT generator [4], [5]. Top-k remained three. The three configurations were C1 (500 characters, 100 overlap; Member 1), C2 (1,200, 200; Member 2), and C3 (800, 100; Member 3). The generation settings were 256 maximum new tokens, temperature 0.2, and sampling enabled. A fresh index was built for each condition.

C. Hardware, Timing, and Scoring

CPU measurements used Colab High-RAM CPU, a compute-unit-consuming option, while GPU measurements used an NVIDIA L4. The initial default free-tier CPU failed to complete even one answer after approximately 40 minutes; it was not used for the reported timed CPU comparison. The notebook reports indexing time as chunk creation plus embedding/index construction, excluding document loading and model initialization [4].

For each configuration-device pair, the group recorded two response-time trials for each of the same nine questions. Response time includes retrieval and generation. The group also assigned one binary correctness/groundedness score per question (1 = correct and supported; 0 = otherwise); the summary reports the number correct out of nine [3]. Mean response time was calculated across the nine per-question means. Speedup = mean CPU response time / mean GPU response time. No separate per-question generation-only timing was logged.

III. EXPERIMENTAL RESULTS

TABLE I. RECORDED CPU/GPU PERFORMANCE FOR THREE RAG CONFIGURATIONS

Dev.	Size	Ovlp.	k	Index (s)	Resp. (s)	Q/9
CPU	500	100	3	17.01	255.83	7/9
L4	500	100	3	2.62	11.06	7/9
CPU	1,200	200	3	17.41	485.94	5/9
L4	1,200	200	3	2.12	10.87	8/9
CPU	800	100	3	15.02	423.46	6/9
L4	800	100	3	2.47	10.88	8/9

Table I reproduces the recorded Part C summary [3]. On the L4 GPU, all three configurations answered within an approximately 10.87-11.06 s mean range. In contrast, the High-RAM CPU means ranged from 255.83 to 485.94 s. The quality assessment is based on the group's recorded binary scores, not an automated factual-verification metric.

TABLE II. GPU RESPONSE-TIME SPEEDUP RELATIVE TO HIGH-RAM CPU

Config.	CPU mean (s)	L4 mean (s)	Speedup
C1 · 500/100	255.83	11.06	23.13x
C2 · 1,200/200	485.94	10.87	44.72x
C3 · 800/100	423.46	10.88	38.91x

C2 recorded the greatest proportional acceleration (44.72x). Its L4 indexing time was 2.12 s, compared with 17.41 s on High-RAM CPU. The difference between the two fastest GPU response means (C2, 10.87 s; C3, 10.88 s) is only approximately 0.02 s and should not be interpreted as a meaningful speed advantage without additional trials.

IV. DISCUSSION

A. Answer Quality by Question Difficulty

The question-level scoring log shows that C2 on L4 correctly answered all three Easy and all three Difficult questions, but only two of the three Average questions. C3 on L4 also achieved 8/9 overall, with three Easy, three Average, and two Difficult questions correct. C1 scored 7/9 on both devices, and its two incorrect responses were the Difficult Moneycat and Tala questions [3].

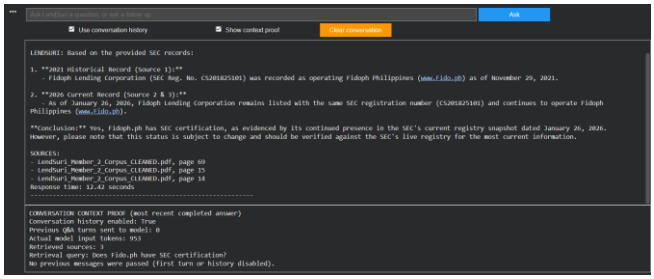
The differences between CPU and GPU correctness for C2 (5/9 versus 8/9) and C3 (6/9 versus 8/9) cannot be attributed to the hardware alone. Generation used sampling, and the workbook does not preserve response texts or per-trial judgments needed to isolate sampling variation, evidence selection, and device-specific effects.

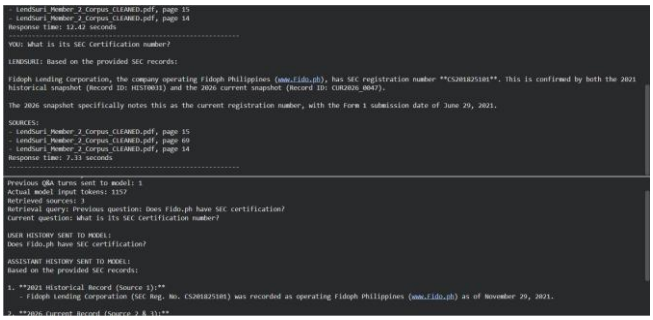
B. Resource Trade-offs and Initial CPU Failure

The initial default Colab CPU attempt was terminated after approximately 40 minutes without a completed answer. The group therefore switched to the compute-unit-consuming High-RAM CPU configuration. Unlike a successful timed response, this incomplete free-tier attempt is a feasibility observation, not a valid numerical speed comparison. The L4 configuration materially reduced observed response times, although its resource cost and session availability were not quantitatively assessed.

V. CONVERSATION-AWARE INTERFACE

After benchmarking, the final notebook added a textbox-driven interactive Q&A interface with session-local memory [4]. The function may pass up to three previous question-and-answer exchanges, limited to 800 history tokens, into the SEA-LION prompt; it also restricts total prompt input to 4,096 tokens. A "Show context proof" diagnostic displays the earlier messages actually included and the token count, and a clear button resets session memory. Retrieved documents remain the primary factual evidence.





This mechanism is present in the submitted notebook; however, no separate recorded follow-up-question accuracy study or conversation-aware timing data appears in the Excel log. Therefore, claims of improved dialogue accuracy or latency are not made, and the Section III performance figures refer only to the original history-free evaluation function.

VI. LIMITATIONS AND REPRODUCIBILITY

The supplied activity asks for at least three measured repetitions, whereas the submitted workbook contains two per question. There is no separately preserved warm-up log, and the workbook lacks independent retrieval/generation times and full generated answers for verifying judgments [1], [3]. Indexing was recorded once per device/configuration, with possible effects from Colab session differences. Further evaluation should add a third or more timed trial, controlled random seeds, response transcripts, and question-level source-grounding assessments.

Reproduction requires the same three unmodified PDFs, nine fixed questions, C1-C3 settings, embedding and generation models, Google Colab notebook, and CPU/GPU hardware details. Source records are dated; absence from a snapshot does not establish unauthorized status, and a listed operator does not authenticate a particular social-media account [2].

VII. RECOMMENDATION AND CONCLUSION

For the tested group workload, C2 (chunk size 1,200; overlap 200; top-k 3) with the L4 GPU is the recommended configuration. It tied for the highest GPU answer-quality result (8/9), achieved all three Difficult questions in the recorded evaluation, and had the lowest recorded GPU indexing time and marginally lowest GPU response mean. C3 (800/100) remains a close alternative with the same overall GPU quality. Where CPU execution is necessary, C1 (500/100) had both the highest recorded CPU quality (7/9) and the lowest CPU response mean (255.83 s). These recommendations apply to the tested dataset and should be rechecked after more repeated trials.

REFERENCES

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